Certifying Safe Operating Regions under Monotone Failure Modes

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Abstract

Many systems are configured by a few continuous knobs, and each setting can fail in more than one way: a language model sampled at low temperature degenerates into repetition and at high temperature into incoherence; a laser powder bed fusion process at low energy leaves unmelted powder and at high energy drills keyhole pores. Practitioners need the *safe operating region*, the settings where every failure mode is rare. Each failure rate can only be estimated from trials at that very setting, so evaluating a grid of configurations costs in proportion to its area. We study *certifying a safe region with a family-wise guarantee when every configuration needs its own trials*, using the one piece of structure these systems share: each failure mode is monotone in known directions. We give (i) the *cone rule*, a closed-testing procedure with betting e-processes that is valid at any stopping time and under any adaptive choice of trials, and under monotonicity reduces to two cumulative sums; (ii) a *search algorithm* for the fixed-confidence problem that stops on its own once the target region is certified; (iii) lower bounds showing that with the ordering the cost is governed by the region's *edge* and without it by its *area*, and an upper bound for the search within logarithmic and multiple-testing factors of the ordered lower bound. On the sampling settings of a language model, where repetition and incoherence rates are measured from real generations, the method certifies half of the safe region with $2.1\times$ fewer generations than testing without the ordering and $3.1\times$ fewer than full-grid Learn-then-Test, pilot study included. Across this, additive manufacturing process windows and a drug-combination screen it needs 1.3–9$\times$ fewer trials than adaptive testing without the ordering and, on 48×48 grids, 19–26$\times$ fewer than a full grid with fixed-sequence testing, with no unsafe setting certified in any run, while a Gaussian-process level-set baseline certifies unsafe settings in 7.5–100% of runs unless it certifies almost nothing.

1. Introduction

A team deploying a language model must choose sampling settings (temperature, top-p). Too low, and outputs repeat themselves; too high, and they stop making sense. They want a region of settings where, with 95% confidence, both failure rates are below 10%, so that any setting in it is safe to ship. A setting's failure rates can only be estimated by generating outputs at that setting and judging them, which is expensive at scale. The same pattern appears in process qualification for additive manufacturing (laser power $\times$ scan speed; lack of fusion vs. keyholing), in dose finding for drug combinations (toxicity vs. lack of response), and in operating envelopes of controllers.

What we ask for. A certified region C such that, with probability at least $1 - \alpha$, every setting in C has every failure rate below its tolerance. The guarantee must hold when the experimenter looks at results after every batch and stops whenever they like, and however the next trials are chosen, including by an imperfect simulator or model.

Why existing tools fall short. *Risk control for configurations* (Learn-then-Test (Angelopoulos et al., 2025); Pareto Testing (Laufer-Goldshtein et al., 2023); Conformal Language Modeling (Quach et al., 2024); MultiRisk (Joshi et al., 2025)) evaluates every candidate configuration on a shared calibration set. When evaluating a configuration means fresh trials at it, that costs the full grid; we compare against Learn-then-Test's fixed-sequence testing, which uses monotonicity along one axis, and find it needs 19–26$\times$ more trials than our method at 48×48. *Adaptive multiple testing with bandits* (Jamieson & Jain, 2018; Xu et al., 2021) samples adaptively but treats configurations as unrelated. *Monotone thresholding bandits* (Garivier et al., 2017; Cheshire et al., 2020) use order in one dimension, for one constraint, without anytime validity. *Safe Bayesian optimization and GP level-set estimation* (Sui et al., 2015; Gotovos et al., 2013; Losalka & Scarlett, 2023; 2024) rely

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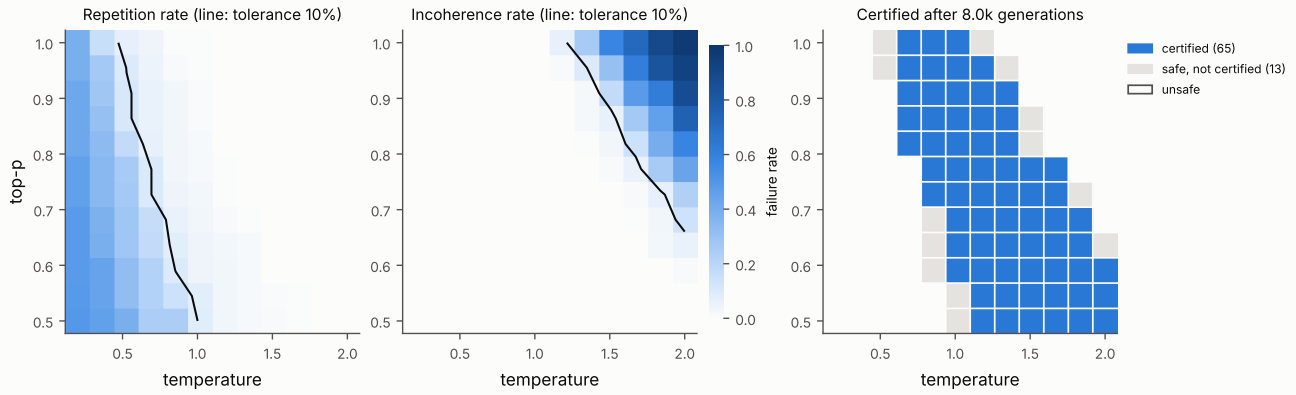


Figure 1. Which sampling settings of a language model are safe? Left and middle: repetition and incoherence rates of SmolLM2-135M, measured from generations over temperature and top-p, with the 10% tolerance (line). Each failure mode is monotone, in opposite directions. Right: the settings our method certifies after 8k generations (about 55 per setting, pilot included), with a 95% guarantee that every certified setting has both rates below 10%. None of them is unsafe. With the same 8k generations, testing without the ordering certifies 50% of the safe region on average and both full-grid baselines certify nothing (55 generations per setting are too few at their per-test levels); ours certifies 83%.

on the GP model being right; their certificates fail when it is not (Section 6).

Structure. In all these systems each failure mode is monotone in known directions (repetition falls with temperature, incoherence rises with it). The null hypotheses "setting x is unsafe for mode m" are therefore logically related: if x is unsafe, so is everything riskier than x.

Contributions.

1. *Problem.* Certification of a safe region under per-mode partial orders when each configuration needs its own trials, in a fixed-budget and a fixed-confidence version (Section 2).

2. *Testing.* The cone rule: closed testing with weighted e-values over exactly the intersections that monotonicity allows, which reduces to a cumulative sum; valid at every time under arbitrary adaptive sampling (Section 3).

3. *Allocation.* For a fixed budget, trials at the riskiest uncertified settings of a slightly shrunken target (the lower bound's binding constraints). For fixed confidence, a galloping/bisection search along the ordering with a two-sided probe that ends almost surely at every failure rate, including rates near the tolerance (Section 4).

4. *Theory.* Lower bounds with and without the ordering (edge vs. area); upper bounds by Wald's identity; the search is within $O(\log n)$ and multiple-testing factors of the ordered lower bound (Section 5).

5. *Experiments.* Sampling settings of a language model, with failure rates measured from real generations (Figure 1), additive manufacturing (a Rosenthal melt-pool model and a synthetic window), and a drug-combination screen (Section 6).

2. Problem

Configurations form a grid $\mathcal{X} = [n_1] \times [n_2]$, $N = n_1 n_2$. There are M failure modes. A trial at x yields $Y \in \{0,1\}^M$, i.i.d. given x (modes may be dependent). The failure rate is $p_m(x) = \mathbb{E}Y_m$; x is *safe* if $p_m(x) < \varepsilon_m$ for all m, and the safe region is W.

Assumption A (monotone failure modes)**.** *For each m there is a known product order $\preceq_m$ on $\mathcal{X}$ such that $x \preceq_m y$ implies $p_m(y) \le p_m(x)$ ("y is at least as safe as x for mode m").*

Protocol. Trials are requested in batches; each batch is chosen from all data so far and any side information (a simulator, a cheaper model, a pilot study) before its outcomes are seen.

Fixed budget. Output certified sets C_t with $\mathbb{P}(C_t \subseteq W \ \forall t) \ge 1 - \alpha$, and make C_t large.

Fixed confidence. For a slack $q < 1$ and target $T = \{x : p_m(x) \le q\varepsilon_m \ \forall m\}$, stop at a time τ with a certified set C.

Definition 1. *A procedure is (α, β)-*correct *if for every instance satisfying Assumption A: (validity) $\mathbb{P}(C \not\subseteq W) \le \alpha$, and (completeness) $\mathbb{P}(\tau < \infty,\ T \subseteq C) \ge 1 - \beta$.*

Settings with rates between $q\varepsilon$ and ε may be certified or not.

3. Testing: the cone rule

Per-configuration evidence. For each (m, x), a betting e-process $K_{m,x}(t) = \prod_{s \le t:\, x_s = x} \big(1 + \lambda_s(\varepsilon_m - Y_{s,m})\big)$, with $\lambda_s \in [0, c/(1-\varepsilon_m)]$, $c < 1$, chosen before trial s. Under $H_{m,x} : p_m(x) \ge \varepsilon_m$ it is a nonnegative supermartingale for the filtration generated by all trials and side information (Lemma 1). We bet with a hedge: half a plug-in bet seeded by side information (Waudby-Smith & Ramdas, 2024), half a mixture of eight fixed bets.

The rule. Fix weights $w_{m,x} \ge 0$ with $\sum w \le 1$ before the trials. Let $\mathcal{C}_m(x) = \{y : y \preceq_m x\}$ be the riskier cone of x (including x). Certify x for mode m once

$$S_m(x,t) = \sum_{y \in \mathcal{C}_m(x)} w_{m,y} K_{m,y}(t) \;\ge\; 1/\alpha, \qquad (1)$$

and certify x once it is certified for every mode. For a product order on a grid, S_m for all x is two cumulative sums.

Theorem 1 (validity)**.** *Under Assumption A, for any allocation and stopping time,*

$$\mathbb{P}\big(\exists\, t, (m,x) :\; p_m(x) \ge \varepsilon_m,\; x \text{ certified for } m \text{ by } t\big) \le \alpha.$$

Proof sketch. The unsafe pairs D are closed toward risk, so a false certification of x gives

$$1/\alpha \le S_m(x,t) \le E(t) := \textstyle\sum_{(m,y)\in D} w_{m,y} K_{m,y}(t),$$

a nonnegative supermartingale starting at most 1; apply Ville's inequality. Full proof in Appendix A.2. □

A certificate at x automatically covers every safer setting, and evidence from riskier settings is pooled. The rule is not an ad hoc pooling scheme: it is exactly closed testing (Marcus et al., 1976) restricted to the intersections that monotonicity allows (Shaffer, 1986), with the weighted e-value intersection test (Vovk & Wang, 2021).

Proposition 1 (closed testing)**.** *Call $I \subseteq [M] \times \mathcal{X}$ admissible if each section $I_m = \{x : (m,x) \in I\}$ is closed toward risk. Under Assumption A the admissible sets are exactly the possible sets of unsafe pairs. Testing each admissible intersection by $\sum_{i \in I} w_i K_i \ge 1/\alpha$ and applying closed testing certifies (m,x) at time t if and only if $S_m(x,t) \ge 1/\alpha$.*

The smallest admissible set containing (m,x) is $\{m\} \times \mathcal{C}_m(x)$, and the intersection statistic grows with I; the proof is in Appendix A.2. Under monotonicity, then, the closure of MN hypotheses costs two cumulative sums instead of a search over 2^{MN} intersections.

Algorithm 1 (search, fixed confidence)

Input: grid, modes and orders, ε, q, α, β, side information.
Probe(m,x): run trials at x until $S_m(x) \ge 1/\alpha$ (certify x and everything safer for m) or $R_{m,x} \ge MN/\beta$ (reject x and everything riskier for m).
For each mode m and row j, riskiest rows first:
  $s \leftarrow$ edge found in an adjacent row, else predicted edge;
  gallop: probe s, then $s \pm 1, s \pm 2, s \pm 4, \ldots$ toward the side the outcome points to, until the outcome flips;
  bisect the undecided positions of the bracket until every position is certified or rejected.
Stop when every (mode, setting) is certified, rejected, or excluded (rejected for another mode); output the certified settings.

Weights. A setting with weight w and rate p is certified after about $\ln(1/(\alpha w)) \,/\, \mathrm{KL}(p\|\varepsilon)$ trials, so the weights should sit on likely edge points. We use the frequency with which a setting is an edge point of the target under perturbed side information (logit shifts), mixed 50/50 with uniform weights.

4. Allocation

Fixed budget: the edge rule. Estimate the target with slack (all estimated rates below 0.7ε); for each mode, run trials at the riskiest uncertified settings of that estimate, several per batch, cheapest first; then re-check uncertified settings whose lower confidence bound still admits membership. These are the binding constraints of the lower bound (Theorem 2a).

Fixed confidence: search (Algorithm 1). For each mode and each row (a chain of the order), run a *two-sided probe* at one setting at a time: certify it by the cone rule, or reject "$p_m(x) \le q\varepsilon_m$" with a failure-betting e-process at level $\beta/(MN)$. A certificate spreads to safer settings, a rejection to riskier ones. Start at the edge found in an adjacent row (or the side information's predicted edge), gallop until the edge is bracketed, then bisect. Stop when every row is decided for every mode. Because $q < 1$, at every rate at least one side of the probe has positive drift; with the fixed bets in our grids this holds at every tolerance we use (condition (T), Section 5), so every probe ends almost surely. A time-out on probe length is also available; it never certifies anything.

5. Theory

Let $K_1 = \mathrm{kl}(1-\beta, \alpha)$, and for $\pi \in [0,1]$ let $f(\pi) = \mathrm{kl}(\pi, 1-\beta)$ for $\pi < 1-\beta$ (else 0) and $g(\pi) = \mathrm{kl}(\pi, \alpha)$ for $\pi > \alpha$ (else 0). Write $\mathrm{KL}_m(y) = \mathrm{KL}(p_m(y)\|\varepsilon_m)$ when $p_m(y) < \varepsilon_m$ and 0 otherwise.

Theorem 2 (lower bound)**.** *Any (α,β)-correct procedure has $\mathbb{E}[\tau] \ge \mathrm{OPT}$, the value of the convex program: mini-*

mize $\sum_y n_y$ *over* $n \geq 0$ *subject to*

(i) for each m *and each riskiest point* t *of* T *(for* $\preceq_m$*):* $\sum_{y \preceq_m t} n_y \mathrm{KL}_m(y) \geq K_1$*;*

(ii) for each m *and each* boundary point z *(*$z \notin T$ *only because of mode* m*, and every setting strictly safer than* z *for* m *is in* T *for* m*):* $(A_z, B_z) \in R$*, where*

$$R = \{(A, B) : \exists \pi \;\; A \geq f(\pi),\; B \geq g(\pi)\},$$

$A_z = n_z \mathrm{KL}(p_m(z) \| q\varepsilon_m)$ *and* $B_z = \sum_{y \preceq_m z} n_y \mathrm{KL}_m(y)$.

Without the ordering, every setting must be resolved on its own and the bound is a sum of per-setting terms over all of $\mathcal{X}$.

The binding points of (i) form an antichain, at most $\min(n_1, n_2)$ per mode, and so do those of (ii). So with the ordering the bound involves at most $2M \min(n_1, n_2)$ settings, and without it all N.

Theorem 3 (oracle upper bound)**.** *Running trials only at a cover of* T *costs at most* $\sum_{cover} (\ln(16/(\alpha w)) + b)/g(p)$ *in expectation, where* $g(p)$ *is the best growth rate of the fixed bets;* $\mathrm{KL}(p\|\varepsilon)/g(p) \in [1.00, 1.05]$ *for* $p \in [0.5\%, 4\%]$ *at* $\varepsilon = 5\%$.

Theorem 4 (search)**.** *Assume the growth condition (T) below. The search is* (α, β)*-correct. If each mode's evidence at* x *uses only the trials of that mode's probes, its expected cost is at most*

$$(C^* + B)\Big(\sum_{m,j} P_{m,j} + (\alpha + \beta)MN\Big),$$

where $P_{m,j} = 2\lceil \log_2 d_{m,j} \rceil + 2$*,* $d_{m,j}$ *is the distance from the first probe in row* j *to the far end of mode* m*'s slack band (rates between* $q\varepsilon_m$ *and* ε_m*),* B *is the batch size, and* $C^* = \max_p \min(T_c(p), T_r(p)) < \infty$ *is the largest expected two-sided probe length.*

Condition (T) asks that at every rate p some fixed certification bet or some fixed rejection bet has positive expected log-growth. It is a property of the bet grids, checked numerically: for ours it holds at every tolerance we use (5–50%) and fails only below about 2%, where adding a smaller rejection bet restores it. The implementation also caps each probe's length (a time-out moves the search on without certifying anything). Time-outs keep validity, make the search stop surely, and cost completeness only through the probability that a target probe times out.

Corollary 1 (edge vs. area)**.** *On an* $n \times n$ *grid the ordered bound involves* $O(Mn)$ *settings and the unordered bound* $O(n^2)$*. The search is within a factor* $O(\log n)$*, the ratio* $C^*/C(p_{boundary})$*, and the multiple-testing factor* $\ln(16MN/(\alpha w))/K_1$ *of the ordered lower bound.*

Whether the $\ln N$ in the multiple-testing factor is necessary for a family-wise guarantee is open; in unstructured pure exploration such terms are necessary in moderate-confidence regimes (Simchowitz et al., 2017).

Proofs are in Appendix A.

6. Experiments

Applications. *Language model sampling.* SmolLM2-135M (Allal et al., 2025); 12 temperatures (0.2–2.0) $\times$ 12 top-p values (0.5–1.0) (Holtzman et al., 2020); prompts from WikiText-2 (Merity et al., 2017); *degenerate* = more than 20% of 4-grams repeated (Welleck et al., 2020), *incoherent* = mean token NLL under the unmodified model above 3.5 nats; $\varepsilon = 10\%$ for both. Ground truth from 144 generations per setting (monotone fit); side information is a pilot of 16 separate generations per setting, counted in the budget. The raw rates are consistent with Assumption A: the largest deviation from the monotone fit is 1.8 standard errors (repetition) and 0.6 (incoherence), and 1 of 264 neighbouring pairs is reversed by more than two standard errors. 78 of the 144 settings are safe (Figure 1). Benchmark runs draw fresh outcomes from the fitted surface. As a check, one live run queried the model directly: with 8.0k real generations it certified 64 of the 78 safe settings and no unsafe one (judged against the measured surface), in line with the 65 certified on average by benchmark runs at the same budget. *Additive manufacturing.* A Rosenthal melt-pool model (Rosenthal, 1946) of 316L with lack-of-fusion (Tang et al., 2017), keyhole (King et al., 2014) and balling criteria, and a synthetic window with controllable simulator error. *Drug combinations.* Toxicity and non-response surfaces with a synergy term; Bliss independence (Bliss, 1939) as side information.

Baselines. *No ordering*: the same e-processes, weights and allocation, without the cone (each setting on its own). *Full grid, Bonferroni*: equal trials everywhere, exact binomial bounds at level $\alpha/(MN)$. *Full grid, fixed sequence*: equal trials everywhere, fixed-sequence testing along each row from the safest end at level $\alpha/(Mn)$, certificates spread to safer settings; this is how Learn-then-Test uses a monotone risk. *GP level set*: a GP classifier surrogate with a confidence-bound certificate.

Fixed budget. Table 1 and Figure 2. The gap to the unordered method grows with resolution (synthetic: 3.2$\times$ at 16^2, 9.1$\times$ at 48^2), as the edge-vs-area bounds predict. The full grids pay for the whole area; fixed-sequence testing helps them by only 20–40%. At 24×24, to certify 70% of the region, ours needs 5.0$\times$ fewer trials than no ordering on the synthetic window, 1.5–1.7$\times$ on Rosenthal 316L and 1.3$\times$ on the drug screen. On the language model, to

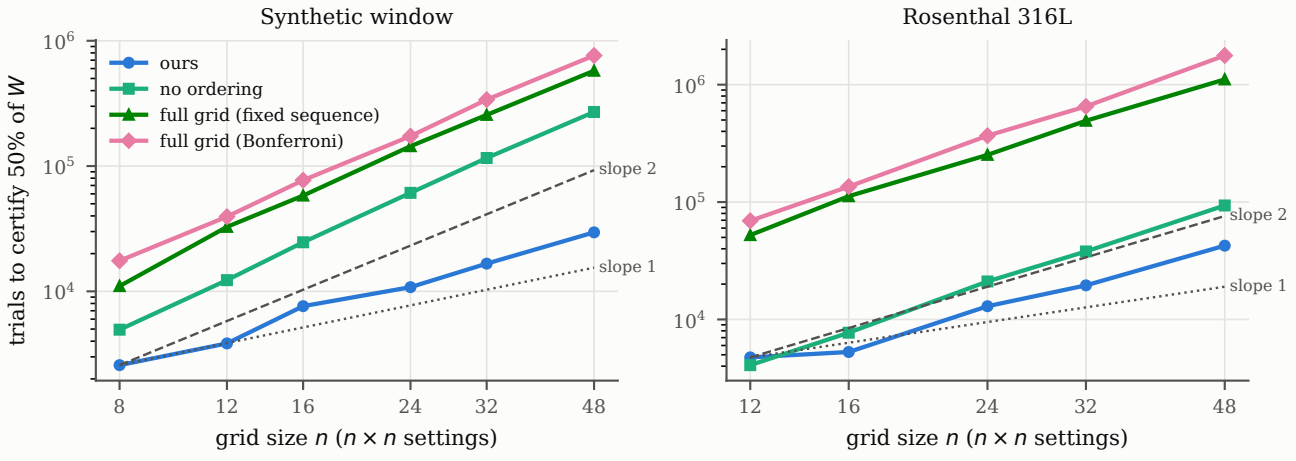


Figure 2. Trials to certify half of the safe region as the grid is refined (fixed budget, log–log). With the ordering the cost grows roughly like the region's edge (slope 1); without it, and for full grids, like its area (slope 2).

Table 1. Trials to certify 50% of the safe region (fixed budget, median of 5 runs, LM 8; full grids: smallest budget reaching 50% on average). LM: generations, with the 2.3k-generation pilot counted for ours and no order.

	grid	ours	no order	grid FST	grid Bonf.
synthetic	16^2	7.6k	24.6k	58.1k	77.1k
	32^2	16.6k	116k	256k	339k
	48^2	29.6k	270k	576k	763k
Rosenthal	16^2	5.3k	7.7k	112k	135k
316L	32^2	19.5k	38.0k	493k	653k
	48^2	42.5k	93.5k	1.11M	1.77M
LM	12^2	3.8k	8.0k	11.7k	18.6k

Table 2. Fixed confidence on the synthetic window (good side information; median of 4 runs) and the language model (median of 8 runs, pilot counted). LB: Theorem 2. The target was fully certified in 23 of 24 synthetic runs (the exception, at 32^2, hit a probe time-out) and in 8 of 8 LM runs.

	with the ordering			without	
grid	search	LB	ratio	per-setting	LB
8^2	14.2k	3.4k	4.2	21.9k	4.5k
16^2	50.3k	8.8k	5.7	100.0k	18.7k
32^2	165.6k	19.4k	8.5	457.7k	77.0k
48^2	334.6k	24.8k	13.5	—	174.4k
LM 12^2	19.7k	2.1k	9.4	31.8k	6.9k

certify 70% of the safe region, ours needs 4.8k generations (pilot included) against 12.7k without the ordering, 20.4k for full-grid fixed-sequence testing and 39.5k for full-grid Bonferroni. Without the pilot it needs 5.3k, so on this small grid the ordering, not the side information, does most of the work. The LM gains do not depend on the failure definitions: re-scoring the same generations with repetition thresholds 0.15–0.3, NLL thresholds 3.0–4.0 nats and tolerances 5–20% (seven settings, 5 runs each) gives 1.6–2.2× fewer generations than no ordering and 2.2–4.1× fewer than full-grid fixed-sequence testing to certify half of the safe region, with no unsafe setting certified in any of the 70 runs.

Fixed confidence. Table 2 and Figure 3. The ordered bound grows roughly linearly in n and the unordered one quadratically, as Theorem 2 predicts. Both methods pay a similar multiple-testing price over their own bound (per-setting: 4.8–5.9×; $\ln(16MN/(\alpha w))/K_1 \approx 5$ at 32^2); the search's extra growth with n is the $O(\log n)$ probe factor. With optimistic side information at 32^2 the search needs 215k trials against 458k without the ordering. On the language model the search stops after 19.7k generations, against 31.8k for the per-setting test, at 9.4 times the ordered lower bound (4.2–5.7 on synthetic grids of similar size), so on this grid the gap to the bound is larger than the gap to the unordered method.

Validity. With every setting exactly at the tolerance (the worst case), the false-certification rate is 0–2.75% across configurations at $\alpha = 5\%$ (400 runs each); no unsafe setting was certified in any efficiency run.

Model-based baseline. The GP level-set certificate certifies an unsafe setting in 40–100% of runs at confidence parameter 2 and still 7.5–20% at 4, where it certifies at most 69% (drug screen) and typically under 25% of the region; ours: 0% and 54–84%. On the language model (20k generations, 20 runs) the GP certifies an unsafe setting in 95–100% of runs at confidence parameter 2 and 15–60% at 3; at 4 it is at 0–10% but certifies 0–78% of the region. Ours, at 40k: 0 of 8 runs, 95% of the region.

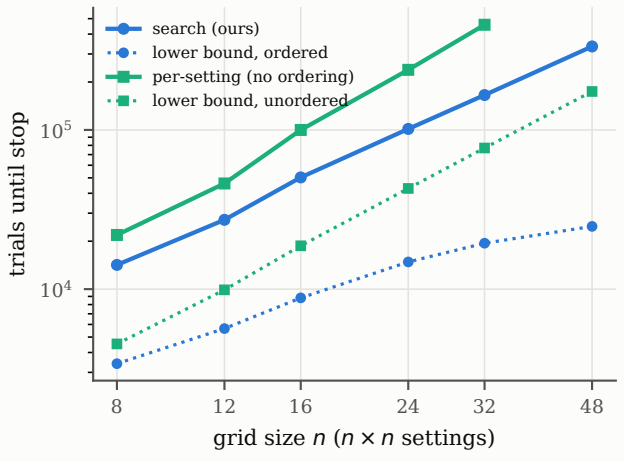


Figure 3. Fixed confidence on the synthetic window: cost at the stopping time and the lower bounds of Theorem 2.

Ablations. The ordering and the allocation rule matter most (no ordering +69 to +213%; greedy allocation +63 to +94%); weights, pooling and bet seeding give smaller gains; a fixed maximum bet fails when relevant rates are near the tolerance.

When monotonicity fails. A pocket of unsafe settings inside the region is certified by spreading in 75–92% of runs; an audit re-testing every certified setting (about 1.2× the budget) catches gross violations (96–100%) but rarely mild ones (4–9%).

7. Related work

Closest work. Three lines come closest. *Closed testing for monotone dose–response designs* (Hellmich & Lehmacher, 2005) exploits the same two-factor order, but with p-values from a fixed design: no adaptive allocation, no optional stopping, no cost analysis. *Learn-then-Test with fixed-sequence testing* (Angelopoulos et al., 2025) uses monotonicity along one parameter on a shared calibration set; on a grid where each setting needs its own trials it is a full-grid method (Table 1). *Structured thresholding bandits* (Garivier et al., 2017; Cheshire et al., 2020; 2021) show that monotonicity changes the sample complexity of classifying arms against a threshold, in one dimension and for a single constraint, with error measured per arm rather than as a family-wise guarantee at any stopping time. We combine the order with e-processes so that validity survives adaptive allocation and stopping, and give bounds showing the edge-versus-area separation for partial orders with several opposing failure modes.

Anytime-valid inference and e-values: Ville (1939); Shafer (2021); Ramdas et al. (2023); Vovk & Wang (2021); Wang & Ramdas (2022); Waudby-Smith & Ramdas (2024). *Multiple testing with structure:* closed testing (Marcus et al., 1976), sequential rejection (Goeman & Solari, 2010), graphical procedures (Bretz et al., 2009), closure for monotone bi-factorial dose designs (Hellmich & Lehmacher, 2005). *Adaptive testing and bandits:* Jamieson & Jain (2018); Xu et al. (2021); thresholding bandits (Locatelli et al., 2016) and their monotone versions (Garivier et al., 2017; Cheshire et al., 2020; 2021); lower bounds (Garivier & Kaufmann, 2016; Kaufmann et al., 2016; Garivier et al., 2019); multi-fidelity best-arm identification (Poiani et al., 2022; 2024). *Risk control for configurations and LLM evaluation:* Angelopoulos et al. (2025); Laufer-Goldshtein et al. (2023); Quach et al. (2024); Joshi et al. (2025); anytime-valid evaluation of a single system (Zhou et al., 2026b;a). *Predictions and simulators inside tests:* prediction-powered inference and e-values (Angelopoulos et al., 2023; Csillag et al., 2025), active inference (Zrnic & Candès, 2024), simulator-guided betting for one robot's mean performance (Chen et al., 2026; Chen & Weng, 2026); we use side information only for allocation and weights, over a whole region. *Safe exploration and level sets:* Sui et al. (2015); Losalka & Scarlett (2023; 2024); Gotovos et al. (2013); Mason et al. (2022). *Applications:* adaptive experimental design for AM process parameters (BEAM authors (Washington State University; incl. J. R. Doppa), 2026); dose finding for combinations (Wages et al., 2011).

8. Limitations

Assumption A is load-bearing (Section 6); thin regions gain less at coarse resolution; the multiple-testing factor dominates the gap to the lower bound; the language-model study uses a small model and automatic judges; the manufacturing studies are simulated with placeholder calibrations.

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A. Proofs

A.1. Setup

Trial $t = 1, 2, \ldots$ is run at x_t and yields $Y_t \in \{0,1\}^M$. Given the past and $x_t = x$, $Y_t \sim P_x$, independently of the past. Let $\mathcal{F}_t = \sigma(x_1, Y_1, \ldots, x_t, Y_t, U)$, where U is independent side information (simulator output, pilot data, internal randomization). The allocation x_t is $\mathcal{F}_{t-1}$-measurable: trials in a batch are chosen before any of them is observed, and processing them one at a time in a fixed order keeps this property. A bet may use earlier trials of the same batch. U must be independent of the trials' outcomes given their settings, which holds when the pilot is a separate set of trials, as in all our experiments.

Lemma 1. *If $p_m(x) \geq \varepsilon_m$ then $K_{m,x}$ is a nonnegative supermartingale with respect to $(\mathcal{F}_t)$. If $p_m(x) \leq q\varepsilon_m$, the rejection process $R_{m,x}(t) = \prod_{s \leq t : x_s = x}(1 + \mu_s(Y_{s,m} - q\varepsilon_m))$, $\mu_s \in [0, 1/(q\varepsilon_m))$ predictable, is a nonnegative supermartingale.*

Proof. Nonnegativity: $1+\lambda(\varepsilon-1) \geq 1-c > 0$. On $\{x_t = x\}$, $\mathbb{E}[K(t) \mid \mathcal{F}_{t-1}] = K(t-1)(1+\lambda_t(\varepsilon - p_m(x))) \leq K(t-1)$, since λ_t and x_t are $\mathcal{F}_{t-1}$-measurable and the conditional law of Y_t given $\mathcal{F}_{t-1}$ and $x_t = x$ is P_x. On $\{x_t \neq x\}$, $K(t) = K(t-1)$. The rejection process is identical with the roles of failures and successes exchanged. A fixed-weight average of such processes (our hedge) is again a nonnegative supermartingale. □

A.2. Validity of the cone rule

Proof of Theorem 1. Let $D = \{(m, y) : p_m(y) \geq \varepsilon_m\}$. Under Assumption A, if $(m, x) \in D$ and $y \preceq_m x$ then $p_m(y) \geq p_m(x) \geq \varepsilon_m$, so $\{m\} \times \mathcal{C}_m(x) \subseteq D$. Let $E(t) = \sum_{(m,y) \in D} w_{m,y} K_{m,y}(t)$, a nonnegative supermartingale by Lemma 1 with $E(0) \leq 1$. A false certification of $(m, x) \in D$ at t means $1/\alpha \leq S_m(x, t) \leq E(t)$. By Ville's inequality $\mathbb{P}(\sup_t E(t) \geq 1/\alpha) \leq \alpha$. □

Proof of Proposition 1. Under A each section of D is closed toward risk (above). Conversely, for admissible I set $p_m = \varepsilon_m$ on I_m and $p_m = 0$ elsewhere; this satisfies A and its unsafe pairs are I. Under H_I the statistic $\sum_I w_i K_i$ is a nonnegative supermartingale starting at most 1, so each intersection test is valid at every stopping time. Closed testing rejects (m, x) when every admissible $I \ni (m, x)$ is rejected. The statistic is nondecreasing in I (nonnegative terms), and $\{m\} \times \mathcal{C}_m(x)$ is admissible (transitivity, other sections empty) and contained in every admissible $I \ni (m, x)$. So all of them are rejected if and only if $S_m(x, t) \geq 1/\alpha$. □

Remarks: nothing about the allocation is used beyond predictability; certification is sticky, and the bound covers "ever certified"; with uniform weights $S_m(x) \geq \max_{y \in \mathcal{C}_m(x)} w K_y$, so the cone rule certifies everything Bonferroni with spreading certifies.

A.3. Correctness of the search

Proof of Theorem 4, correctness. Validity is Theorem 1. Completeness: let G be the event that no (m, x) with $p_m(x) \leq q\varepsilon_m$ is rejected directly. By Lemma 1, Ville and a union bound over MN pairs, $\mathbb{P}(G^c) \leq \beta$. On G no $x \in T$ is rejected or excluded: a spread rejection of x for m comes from a direct rejection at some y with $x \preceq_m y$, but then $p_m(y) \leq p_m(x) \leq q\varepsilon_m$, contradicting G; exclusion through another mode is the same argument for that mode. Termination: let $g_c(p) = \max_j \mathbb{E}_p \ln(1 + \lambda_j(\varepsilon - Y))$ and $g_r(p) = \max_k \mathbb{E}_p \ln(1 + \mu_k(Y - q\varepsilon))$ over the fixed bets. Condition (T) is $\max(g_c(p), g_r(p)) > 0$ for all $p \in [0, 1]$; for our grids at $q = 0.5$ the minimum over p is $1.6 \cdot 10^{-3}$, $4.6 \cdot 10^{-3}$, $1.6 \cdot 10^{-2}$ and $3.4 \cdot 10^{-2}$ at $\varepsilon = 5, 10, 30, 50\%$. If $g_c(p) > 0$, the strong law gives $\frac{1}{n} \ln K_j \to g_c(p) > 0$ for the best fixed bet, and $S_m(x) \geq w_{m,x} K_j / 16$ crosses $1/\alpha$ a.s.; if $g_r(p) > 0$, the rejection mixture ($\geq R_k/5$) crosses its level a.s. So every probe ends a.s., each decides at least one of the MN pairs, and $\tau < \infty$ a.s. So on G, $\tau < \infty$ and $T \subseteq C$. A time-out marks a pair rejected for the search only; it never certifies, so validity holds, and completeness holds on G intersected with the event that no target probe times out. □

A.4. Expected passage times

Lemma 2 (Wald)**.** *Let $Z_k = \sum_{i \leq k} \xi_i$ with ξ_i i.i.d., bounded, $\mathbb{E}\xi = g > 0$, and $\tau = \min\{k : Z_k \geq c\}$. Then $\mathbb{E}\tau \leq (c + b)/g$, $b = \operatorname{ess\,sup} \xi$.*

Proof. $\mathbb{E}\tau < \infty$ for positive drift with bounded increments; by Wald's identity $\mathbb{E}Z_\tau = g\mathbb{E}\tau$, and $Z_\tau \le c + b$. □

Applied to the log-wealth of the best fixed bet in the hedge (held with weight $1/16$) and $S_m(x) \ge w_{m,x}K_{m,x}$: a setting with rate $p < \varepsilon$ sampled exclusively is certified within an expected $T_c(p) = (\ln(16/(\alpha w)) + \ln(1 + \lambda^*\varepsilon))/g_c(p)$ trials, where $g_c(p) = \max_j \mathbb{E}\ln(1 + \lambda_j(\varepsilon - Y))$. Likewise rejection at $p > q\varepsilon$ within $T_r(p) = (\ln(5MN/\beta) + b')/g_r(p)$. This gives Theorem 3 by summing over the cover (plus B per cover point when certification is checked once per batch).

Lemma 3 (probes per row)**.** *Fix m and a row ordered safest to riskiest for m; let a be the last position with $p_m \le q\varepsilon_m$ and b the first with $p_m \ge \varepsilon_m$. On the event that no e-process misleads (probability at least $1 - \alpha - \beta$), a probe at i can certify only if $i < b$ and reject only if $i > a$, and if the first probe is at s the row is decided after at most $2\lceil \log_2 d \rceil + 2$ probes, $d = \max(b - s, s - a, 1)$.*

Proof. If the first probe certifies, galloping probes $s + 1, s + 2, s + 4, \ldots$ and cannot certify at or beyond b, so it rejects within $\lceil \log_2(b - s) \rceil + 1$ probes, leaving at most $b - s$ undecided positions in the bracket. A probe at the median of the undecided positions certifies it and everything safer or rejects it and everything riskier, halving them. Positions decided elsewhere only shrink the bracket. The other case is symmetric. □

Proof of Theorem 4, cost. Number the probes in the order they start. Probes interleave, so use the reward-table construction: for each (m, x) draw in advance an i.i.d. sequence from P_x, and let the j-th trial of a mode-m probe at x read its j-th entry. This has the procedure's law. In this variant each pair is probed at most once (a probe ends only when its pair is decided). So when probe k starts at (m, x), its processes start fresh and the trials it will see at x are i.i.d. and independent of the past; since $S_m(x) \ge w_{m,x}K_{m,x}$, it ends no later than the first of the two passage times of Lemma 2, so its conditional expected length is at most $\min(T_c, T_r) + B \le C^* + B$ (B for checking once per batch). Whether probe k happens is known when it starts, so the expected total is at most $(C^* + B)$ times the expected number of probes. By Lemma 3 that number is at most $\sum_{m,j} P_{m,j}$ on the good event, and at most MN otherwise. □

A.5. Lower bounds

We use the transportation lemma (Kaufmann et al., 2016, Lemma 1): for instances ν, ν', a stopping time τ and $E \in \mathcal{F}_\tau$, $\sum_x \mathbb{E}_\nu[N_x(\tau)]\mathrm{KL}(\nu_x, \nu'_x) \ge \mathrm{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E))$. We need $\nu_x \ll \nu'_x$ for every x (both our alternatives keep this, including at $p = 0$) and $E \subseteq \{\tau < \infty\}$ (true of our events). It needs only $\mathbb{E}_\nu[\tau] < \infty$ (otherwise there is nothing to prove): then $\tau < \infty$ ν-a.s., the restriction of $\mathbb{P}_\nu$ to $\mathcal{F}_\tau$ is absolutely continuous with respect to that of $\mathbb{P}_{\nu'}$ with density the likelihood ratio at τ, Wald's identity gives the left side as the expected log-likelihood ratio, and the data-processing inequality gives the right side; τ may be infinite under ν'. If $\mathbb{P}_{\nu'}(E) \le \alpha$ the right side is at least $g(\mathbb{P}_\nu(E))$, and if $\mathbb{P}_{\nu'}(E) \ge 1 - \beta$ it is at least $f(\mathbb{P}_\nu(E))$. We restrict to instances with independent modes (a lower bound on a subclass is a lower bound), so KL adds over modes, and a trial at y counts once in every constraint that involves y.

Proof of Theorem 2. (i) For m and t, let ν' set $p'_m(y) = \max(p_m(y), \varepsilon_m + \eta)$ for $y \preceq_m t$ and leave everything else unchanged. ν' satisfies Assumption A: the riskier cone of t is closed toward risk, and raising a monotone function to at least a constant on such a set keeps it monotone (check the three cases of a comparable pair). In ν', t is unsafe, so $\mathbb{P}_{\nu'}(T \subseteq C) \le \alpha$, while $\mathbb{P}_\nu(\tau < \infty, T \subseteq C) \ge 1 - \beta$; apply the lemma and let $\eta \to 0$.

(ii) Lowering alternative: $p''_m(z) = q\varepsilon_m$, everything else unchanged. This keeps Assumption A because every setting strictly safer than z has $p_m \le q\varepsilon_m$ and every riskier one has $p_m \ge p_m(z) > q\varepsilon_m$. In ν'', $z \in T$, so $\mathbb{P}_{\nu''}(z \in C) \ge 1 - \beta$, giving $A_z \ge f(\pi)$ with $\pi = \mathbb{P}_\nu(z \in C)$. Raising alternative: as in (i) with t replaced by z; z is unsafe there, giving $B_z \ge g(\pi)$ for the same π.

R is convex: its lower-left boundary is $B = g(f^{-1}(A))$ on the range of f, an increasing convex function composed with a decreasing convex one. Replacing R by finitely many supporting half-planes $wA + (1-w)B \ge \min_\pi[wf(\pi) + (1-w)g(\pi)]$ gives a linear program with a smaller value, which is what we report.

Without the ordering, alternatives may change a single setting x, so the constraints decouple. With

$$R_x = \sum_{m:\, p_m(x) > q\varepsilon_m} \mathrm{KL}(p_m(x) \| q\varepsilon_m), \qquad C_x = \min_m \mathrm{KL}(p_m(x) \| \varepsilon_m),$$

$$n_x \;\geq\; \begin{cases} K_1/C_x, & x \in T, \\ \min_\pi \max\big(f(\pi)/R_x,\; g(\pi)/C_x\big), & x \notin T,\; x \text{ safe}, \\ \mathrm{kl}(\alpha, 1-\beta)/R_x, & x \text{ unsafe}. \end{cases}$$

□